

* Echo	Format	Session Time	Transport Layer	SIP: DIP
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② Data Link Layer:-

- * In this layer using Ethernet Protocol 802.3 and MAC is Added.

* Echo	Format	Session Time	Transport Layer	SIP:- DIP:-	S MAC D MAC
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- * So if in case the Destination MAC is not Available we can find it with the help of ARP

ARP → Address Resolution Protocol

- * It is used to find the Destination MAC with the help of Destination IP

ARP Request Packet:-

My IP:- 50.0.0.10
My MAC:- 00A
Tell me who is the DIP → 50.0.0.20
Give your MAC

So in this Request packet the values will be

SIP:- 50.0.0.10
S MAC:- 00A
Target IP:- 50.0.0.20
Target MAC:- FFF

So, S MAC:- 00A

D MAC:- FFF → Broadcast MAC

ARP Reply Packet

My IP:- 50.0.0.20
My MAC 00B
Your IP 50.0.0.10
Your MAC 00A

So in this Reply packet will be

Sender IP:- 50.0.0.20
Sender MAC:- 00B
Target IP:- 50.0.0.10
Target MAC:- 00A

So, S MAC:- 00B

D MAC:- 00A

IPs with corresponding DIP will read and they
 whereas the other IP devices will open and close

- * Note: - Unlike decapsulation the MAC is present first
 So all the devices will open, After seeing
 the IP it will drop.

* So ARP Request is a Broadcast message.

So in ARP Reply:-

- * It is a unicast message
- * As all the MAC and IP is known it will send the message

After receiving ARP Reply:-

- * The MAC address is noted down in a Table
- * The Table validity is for 5mins
- * Each time the data is used it will be renewed again 5min.

ARP Table	
IP	MAC
50.0.0.10	00A
50.0.0.20	00B

So after getting MAC the echo data is sent via the

① Physical Layer → NIC

Echo SIP:- 50.0.0.10 SMAC:- 00A
 DIP:- 50.0.0.20 DMAC:- 00B →

← Echo SIP:- 50.0.0.20 SMAC:- 00B
 DIP:- 50.0.0.10 DMAC:- 00A